S.No. : 335

BCAN15324

No. of Printed Pages : 04

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 51133 Roll No. 1 2 3 0 2 1 1 1 9 9

BCA Examination 2025-26

Odd Semester

MACHINE LEARNING

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : $1 \times 8 = 8$

   (a) Define machine learning.

   (b) What is the difference between supervised and unsupervised learning?

   (c) What is overfitting in machine learning?

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(d) Define training data and testing data.

(e) What is role of a cost function in ML models?

(f) Define classification and regression.

(g) What is confusion matrix?

(h) Mention two applications of machine learning in real life.

SECTION – B

2. Attempt any two parts of the following . $2 \times 6 = 12$

(a) Discuss various applications of machine learning.

(b) Differentiate between batch learning and online learning with examples.

(c) Explain K-Nearest neighbor algorithm with a suitable example.

(d) Explain CNN architecture and its different layers.

SECTION – C

Note:- Attempt any two parts from each question. Each part carry equal marks. $8 \times 5 = 40$

3. (a) Explain the working of decision tree algorithm with an example.

   (b) What are the advantages and disadvantages of support vector machines?

   (c) Explain the Naive Bayes classifier and its assumptions.

4. (a) Describe linear regression and derive its cost function.

   (b) Write a short note on Q-learning and Markov decision process.

   (c) Explain dimensionality reduction using principal component analysis (PCA).

5. (a) What are ensemble methods? Explain bagging and boosting?

(b) Explain clustering techniques in machine learning with examples.

(c) Discuss different model evaluation metrics used in classification problems.

6. (a) Explain gradient descent and its types.

(b) Write a short note on reinforcement learning.

(c) What is an ANN? Explain its applications.

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